

3008ICT Lab 9 (Attention and Transformers)

Task 1. Questions and Discussions

Part 1: True / False

Instructions: Indicate whether the following statements are True or False based on the concepts of Deep Learning, Transformers, and Attention.

1. In a self-attention layer, the input is copied to act as the key, the value, and the query.
2. Layer normalisation in a Transformer block calculates the mean and variances along the first batch dimension, exactly like standard batch normalisation.
3. In an attention layer, the output is generated by computing a weighted sum of the values.

Part 2: Multiple Choice

Instructions: Select the best answer for each question.

1. What is the primary motivation for introducing the Attention mechanism in sequence-to-sequence models?

- A) To compress all source tokens into a single hidden state vector.
- B) To replace all dense layers in the encoder.
- C) To allow each generated token to relate to different source tokens rather than relying on a single hidden state.
- D) To reduce the number of parameters in the network.

2. Which of the following components is used to inject sequential information into a Transformer model?

- A) Recurrent layers
- B) Positional Encoding
- C) Position-wise Feed-Forward Networks
- D) Multi-head Attention

3. How does a Position-wise Feed-Forward Network (FFN) in a Transformer reshape the input data?

- A) From (seq_len, fea_size) to (batch, seq_len)

- B) From (batch, seq_len, fea_size) to (batch * seq_len, fea_size)
- C) From (batch, fea_size) to (batch * seq_len, fea_size)
- D) From (batch * seq_len, fea_size) to (batch, seq_len, fea_size)

Part 3: Short Answer

Instructions: Provide a concise explanation for the following prompts.

1. **Calculating Attention:** Briefly explain the mathematical difference between how "Dot Product Attention" and "Multilayer Perceptron (MLP) Attention" calculate their attention scores.
2. **Positional Encoding:** Why does a Transformer model require Positional Encoding, and mathematically, what types of functions are used to create the positional matrix P ?

Part 4: Long Answer / Conceptual

Instructions: Provide a detailed response demonstrating your understanding of the architectures.

1. **Architectural Differences:** Compare and contrast a traditional Seq2Seq model equipped with attention versus the pure Transformer architecture. Identify and describe the three major places where the Transformer differs from a Seq2Seq model with attention.

Sample Answers (for some questions, one can come up with slightly different answers):

Part 1: True / False

1. **True**
2. **False** (It calculates the mean and variances along the last dimension/feature dimension, not the first batch dimension).
3. **True**

Part 2: Multiple Choice

1. C

2. B

3. B

Part 3: Short Answer

1. **Dot Product Attention** calculates scores using a vectorised dot product of queries and keys scaled by the square root of the dimension size:

$$\alpha(Q, K) = QK^T / \sqrt{d}.$$

MLP Attention concatenates the key and query, applies learnable weight matrices, and feeds them into a single hidden-layer perceptron using a *tanh* activation function.

2. Transformers run in parallel and do not preserve sequential information by default. Positional encoding injects this missing sequential context. The positional matrix P is created using *sine* and *cosine* functions of different frequencies.

Part 4: Long Answer

1. Key Differences:

- The Transformer relies entirely on attention and positional encoding, dropping recurrent layers entirely.
- It utilises stacked "Transformer blocks" that contain Multi-head Attention, Add & Norm (Layer Normalisation), and Position-wise FFNs.
- The state from the encoder is passed to *every* block in the decoder, rather than just acting as a single hidden state passed between recurrent layers.